

1 Idea: Neural Observation Encoder + Temporal Graphical Inference

1.1 Goal

Given a video with T frames, we first extract frame-level visual observations from each frame, and then infer a structured latent temporal trajectory. The inferred hidden trajectory is finally used for motion classification.

Formally, let

$$X_{1:T} = (X_1, X_2, \dots, X_T)$$

denote the frame-level inputs, where X_t is the token representation of frame t . Our goal is to infer a latent trajectory

$$Y_{1:T} = (Y_1, Y_2, \dots, Y_T), \quad Y_t \in R^d,$$

and use it to predict a motion label

$$C \in \mathcal{C}.$$

1.2 Step 1: Shared Neural Encoder for Local Evidence

For each frame t , we use a shared encoder to map the observed frame tokens X_t into a frame-level latent proposal:

$$Z_t = f_\theta(X_t), \quad t = 1, \dots, T.$$

Here:

- X_t is the token representation of frame t ,
- f_θ is a shared neural encoder applied to every frame,
- $Z_t \in R^d$ is a frame-wise local evidence point.

The role of Z_t is not to be the final hidden state itself, but rather to provide local observation evidence for the latent temporal variable Y_t .

1.3 Step 2: Temporal Graphical Model over Hidden States

We introduce hidden variables $Y_1, \dots, Y_T$ and impose a chain-structured temporal dependency:

$$Y_1 \leftrightarrow Y_2 \leftrightarrow \dots \leftrightarrow Y_T.$$

Conditioned on the observation evidence $Z_{1:T}$, we define the hidden trajectory distribution as

$$p(Y_{1:T} \mid Z_{1:T}) \propto \prod_{t=1}^T \phi_{\text{obs}}(Y_t, Z_t) \prod_{t=2}^T \psi_{\text{trans}}(Y_{t-1}, Y_t).$$

We use the following observation factor:

$$\phi_{\text{obs}}(Y_t, Z_t) = \exp\left(-\frac{1}{2\sigma_x^2}\|Y_t - Z_t\|^2\right),$$

which encourages the inferred hidden state Y_t to stay close to the frame-wise local evidence Z_t .

We use the following transition factor:

$$\psi_{\text{trans}}(Y_{t-1}, Y_t) = \exp\left(-\frac{1}{2\sigma_t^2}\|Y_t - AY_{t-1}\|^2\right),$$

where $A \in R^{d \times d}$ is a learnable transition matrix. This factor enforces temporal consistency of the hidden trajectory.

A simpler version is

$$\psi_{\text{trans}}(Y_{t-1}, Y_t) = \exp\left(-\frac{1}{2\sigma_t^2}\|Y_t - Y_{t-1}\|^2\right),$$

which directly encourages adjacent hidden states to be close.

1.4 Step 3: MAP Inference of the Hidden Trajectory

Instead of directly taking Z_t as the final representation, we infer the latent trajectory by MAP inference:

$$Y_{1:T}^* = \arg \max_{Y_{1:T}} p(Y_{1:T} \mid Z_{1:T}).$$

Equivalently, this is the minimizer of the following energy:

$$Y_{1:T}^* = \arg \min_{Y_{1:T}} E(Y_{1:T}),$$

where

$$E(Y_{1:T}) = \sum_{t=1}^T \|Y_t - Z_t\|^2 + \lambda \sum_{t=2}^T \|Y_t - AY_{t-1}\|^2.$$

In the simpler case with identity transition,

$$E(Y_{1:T}) = \sum_{t=1}^T \|Y_t - Z_t\|^2 + \lambda \sum_{t=2}^T \|Y_t - Y_{t-1}\|^2.$$

Thus, the neural encoder provides local frame-wise evidence, while the graphical model refines these local proposals into a globally consistent temporal hidden trajectory.

1.5 Step 4: Classification from the Inferred Trajectory

After inferring $Y_{1:T}^*$, we use the hidden trajectory to predict the motion label:

$$\hat{C} = g_\omega(Y_{1:T}^*),$$

where g_ω is a lightweight classifier.

For example, we may define

$$s = [Y_1^*, Y_T^*, Y_T^* - Y_1^*, \bar{Y}^*], \quad \bar{Y}^* = \frac{1}{T} \sum_{t=1}^T Y_t^*,$$

and then predict

$$\hat{C} = g_\omega(s).$$

1.6 Training Objective

We train the model using a classification loss together with the structured energy term:

$$\mathcal{L} = \text{CE}(\hat{C}, C) + \beta E(Y_{1:T}^*).$$

Here:

- $\text{CE}(\hat{C}, C)$ is the cross-entropy classification loss,
- $E(Y_{1:T}^*)$ is the structured temporal energy,
- $\beta > 0$ controls the strength of temporal regularization.

1.7 Interpretation

This model has two conceptually distinct parts:

1. A **shared neural encoder** f_θ that converts each frame into a local observation evidence point Z_t ;
2. A **temporal graphical model** that infers a globally consistent hidden trajectory $Y_{1:T}$ under observation and transition constraints.

Therefore, the model is not merely learning frame-wise embeddings. Instead, it performs structured latent-state inference over time, which is the main connection to probabilistic graphical models.

1.8 Why This Is Related to PGM

The relation to probabilistic graphical models comes from the following aspects:

- There is an explicit sequence of hidden variables $Y_1, \dots, Y_T$;
- These hidden variables are connected by a chain structure;

- The model factorizes into observation terms and transition terms;
- The final hidden trajectory is obtained by structured inference rather than direct feed-forward prediction.

Hence, this project can be viewed as a chain-structured latent graphical model conditioned on neural frame-level observations.